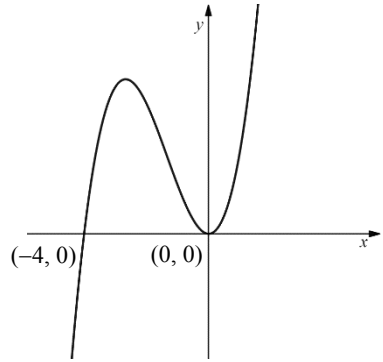
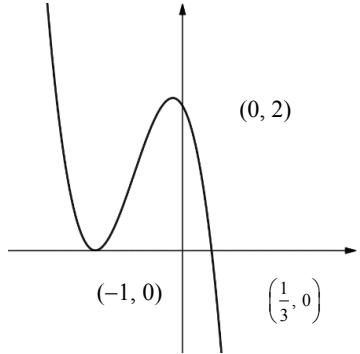
1	$\int\left(10x^4-\frac{3}{2x^3}-7\right)\mathrm{d}x=2x^5+\frac{3}{4}x^{-2}-7x+c$	M1A1A1
		(3)
		Total 3

2(i)(a)	$2^{n+3} = 2^n \times 2^3 = 8m$	B1
		(1)
(b)	$16^{3n} = \left(2^4\right)^{3n}$	M1
	$= 2^{12n} = \left(2^n\right)^{12} = m^{12}$	A1
		(2)

(ii)	$x\sqrt{3}-3=x+\sqrt{3}\Rightarrow x\sqrt{3}-x=3+\sqrt{3}\Rightarrow x\left(\sqrt{3}-1\right)=3+\sqrt{3}\Rightarrow x=...$	M1
	$=\dfrac{3+\sqrt{3}}{\sqrt{3}-1}\times\dfrac{\pm\left(\sqrt{3}+1\right)}{\pm\left(\sqrt{3}+1\right)}$	M1
	$=\dfrac{\pm\left(4\sqrt{3}+6\right)}{\pm\left(3-1\right)}=3+2\sqrt{3}$	A1
		(3)

(ii) ALT 1	$x\sqrt{3}-3=x+\sqrt{3}\Rightarrow\left(p+q\sqrt{3}\right)\sqrt{3}-3=p+q\sqrt{3}+\sqrt{3}\Rightarrow p\sqrt{3}+3q-3=p+\sqrt{3}+q\sqrt{3}$ $\Rightarrow p=q+1,\ 3q-3=p$	M1
	$\Rightarrow p=q+1,\ 3q-3=p\Rightarrow p=...,q=...$	M1
	$x=3+2\sqrt{3}$	A1
(ii) ALT 2	$x\sqrt{3}-3=x+\sqrt{3}\Rightarrow 3x^2-6\sqrt{3}x+9=x^2+2\sqrt{3}x+3$ $\Rightarrow 2x^2-8\sqrt{3}x+6=0$	M1
	$x^2-4\sqrt{3}x+3=0\Rightarrow x=\dfrac{4\sqrt{3}\pm\sqrt{\left(4\sqrt{3}\right)^2-4\times1\times3}}{2}$	M1
	$\Rightarrow x=\dfrac{4\sqrt{3}\pm\sqrt{36}}{2}\Rightarrow x=2\sqrt{3}\pm3$ $x=2\sqrt{3}+3$	A1
(ii) ALT 3	$x\sqrt{3}-3=x+\sqrt{3}\Rightarrow\left(x\sqrt{3}-3\right)\left(x-\sqrt{3}\right)=\left(x+\sqrt{3}\right)\left(x-\sqrt{3}\right)$ $\Rightarrow\sqrt{3}x^2-6x+3\sqrt{3}=x^2-3\Rightarrow\left(\sqrt{3}-1\right)x^2-6x+3\sqrt{3}+3=0$	M1
	$\left(\sqrt{3}-1\right)x^2-6x+3\sqrt{3}+3=0\Rightarrow x=\dfrac{6\pm\sqrt{36-4\left(\sqrt{3}-1\right)\left(3\sqrt{3}+3\right)}}{2\left(\sqrt{3}-1\right)}$ $=\dfrac{6\pm\sqrt{36-24}}{2\left(\sqrt{3}-1\right)}=\dfrac{6\pm\sqrt{12}}{2\left(\sqrt{3}-1\right)}=\dfrac{3\pm\sqrt{3}}{\sqrt{3}-1}=\dfrac{3\pm\sqrt{3}}{\sqrt{3}-1}\times\dfrac{\sqrt{3}+1}{\sqrt{3}+1}$	M1
	$=\dfrac{3\sqrt{3}+3\pm3\pm\sqrt{3}}{2}=2\sqrt{3}+3,\ \sqrt{3}$ $x=2\sqrt{3}+3$	A1
		Total 6

3(a)		B1B1B1
		(3)

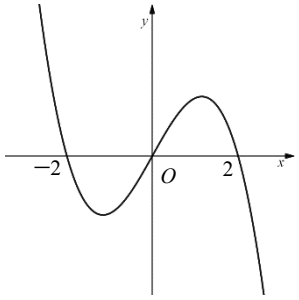
(b)		B1B1B1
		(3)
		Total 6

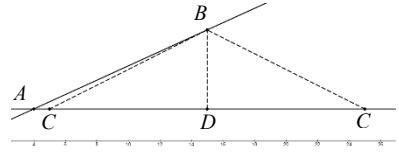
4(a)	$x^2 + kx - 9 = -3x^2 - 5x + k \Rightarrow 4x^2 + kx + 5x - 9 - k\,(= 0)$	M1
	$b^2 - 4ac = 0 \Rightarrow (k+5)^2 - 4 \times 4(-9-k) = 0$	M1
	$k^2 + 26k + 169 = 0$*	A1*
		(3)
(b)	$k^2 + 26k + 169 = 0 \Rightarrow k = -13$	B1
	$k = -13 \Rightarrow 4x^2 - 8x + 4 = 0 \Rightarrow x = ...$	M1
	$(1, -21)$	A1
		(3)
		Total 6

5(a)	$\frac{1}{2}\times 6^2\times 1.3=...$	M1
	$=23.4\,(\text{m}^2)$	A1
		(2)
(b)	$12.2^2=6^2+10.8^2-2\times 6\times 10.8\cos(ABE)$	M1
	$\cos(ABE)=\frac{6^2+10.8^2-12.2^2}{2\times 6\times 10.8}\left(=\frac{19}{648}\right)$ $ABE=1.54$	A1
		(2)
(c)	Area $ABE=\frac{1}{2}\times 10.8\times 6\sin(ABE)$	M1
	Area $BCD=\frac{1}{2}\times 6\cos(\pi-1.3-"1.54")\times 6\sin(\pi-1.3-"1.54")$ or e.g. Area $BCD=\frac{1}{2}\times 6\sin(\pi-1.3-"1.54")\times\sqrt{6^2-(6\sin(\pi-1.3-"1.54"))^2}$	M1
	Total area $=60.9\text{m}^2$	A1
		(3)
		Total 7

6(a)	$y-5x=75,\ y=2x^2+x-21$ $\Rightarrow 2x^2+x-21=5x+75$ $\Rightarrow 2x^2-4x-96=0$ or e.g. $x^2-2x-48=0$ or $\Rightarrow y=2\left(\frac{y-75}{5}\right)^2+\frac{y-75}{5}-21$ $\Rightarrow 2y^2-320y+10350=0$ or e.g. $y^2-160y+5175=0$	M1
	$x^2-2x-48=0\Rightarrow (x-8)(x+6)=0\ \Rightarrow x=-6,\ 8$	dM1
	$x=-6\Rightarrow y=45$ or $x=8\Rightarrow y=115$	dM1
	$P(-6,\ 45)$ and $Q(8,\ 115)$	A1
		(4)
(b)	e.g. $y\leqslant 2x^2+x-21,\ y-5x\leqslant 75,\ y\geqslant 0,\ x\leqslant -3.5$ $x\leqslant a$ where $-3.5\leqslant a<3$ $(\text{or } a\leqslant x\leqslant b \text{ where } a\leqslant -15, -3.5\leqslant b\leqslant 3)$	M1A1A1
		(3)
		Total 7

7(a)(i)	$f(x)=2x^3-kx^2+14x+24 \Rightarrow (f'(x)=)6x^2-2kx+14$	M1A1
	$(f''(x)=)12x-2k$	A1ft
(ii)		**(3)**
(b)	$6x^2-2kx+14=12x-2k \Rightarrow 6(5)^2-2k(5)+14=12(5)-2k \Rightarrow k=...$	M1
	$k=13$	A1
		(2)
(c)	$k=13 \Rightarrow 6x^2-38x+40=0 \Rightarrow x=...$	M1
	$x="\frac{4}{3}" \Rightarrow y=12\left("\frac{4}{3}"\right)-2\times13$ or $y=6\left("\frac{4}{3}"\right)^2-2\times13\times\left("\frac{4}{3}"\right)+14$	M1
	$x=\frac{4}{3},\ y=-10$	A1
		(3)
		Total 8

8(a)		B1B1B1
		(3)
(b)	$x\left(4-x^2\right)=\dfrac{A}{x}\Rightarrow 4x^2-x^4=A$ $\Rightarrow x^4-4x^2+A=0*$	B1*
		(1)
(c)	$A>0$	B1
	$b^2=4ac\Rightarrow 16=4A\Rightarrow A=...$	M1
	$0<A<4$	A1
		(3)
		Total 7

9(a)	$m=\dfrac{7-2}{15-4}\left(=\dfrac{5}{11}\right)$	M1
	$y-2=\dfrac{5}{11}(x-4)$ or $y-7=\dfrac{5}{11}(x-15)$ or $y=\dfrac{5}{11}x+c \Rightarrow 2=\dfrac{5}{11}\times 4+c \Rightarrow c=...\left(\dfrac{2}{11}\right)$	M1
	$5x-11y+2=0$	A1
		(3)
(a) ALT	$y=mx+c \Rightarrow \begin{matrix}2=4m+c\\7=15m+c\end{matrix}$	M1
	$\begin{matrix}2=4m+c\\7=15m+c\end{matrix} \Rightarrow m=...\left(\dfrac{5}{11}\right),\quad c=...\left(\dfrac{2}{11}\right)$	M1
	$5x-11y+2=0$	A1
(b)	$(15-x)^2+(7-2)^2=\left(5\sqrt{5}\right)^2$ or e.g. $\sqrt{(15-x)^2+(7-2)^2}=5\sqrt{5}$ or e.g. $\left(5\sqrt{5}\right)^2=5^2+CD^2$ oe 	M1
	Way 1: $(15-x)^2+25=125 \Rightarrow (15-x)^2=100 \Rightarrow x=...$ or e.g. Way 2: $125=25+CD^2 \Rightarrow CD^2=100 \Rightarrow CD=10 \Rightarrow x=...$	M1
	$(5,\ 2)$ **or** $(25,\ 2)$	A1
	$(5,\ 2)$ **and** $(25,\ 2)$	A1
		(4)
(c)	Area $=\dfrac{1}{2}("5"-4)\times 5$	M1
	$=\dfrac{5}{2}$	A1
		(2)
		Total 9

10(a)	$\mathrm{f}'(4)=6(4)-\dfrac{7\times14}{4}=-\dfrac{1}{2}$	B1
	$m_T=-\dfrac{1}{2}\Rightarrow m_N=\dfrac{-1}{-1/2}$	M1
	$y-12=2(x-4)$ or $y=mx+c\Rightarrow y=2x+c\Rightarrow 12=2(4)+c\Rightarrow c=...$	M1 (A1 on ePen)
	$y=2x+4$	A1
		(4)
(b)	$...\dfrac{(2x-1)(3x+2)}{2\sqrt{x}}=...\dfrac{6x^2+x-2}{2\sqrt{x}}=...3x^{\frac{3}{2}}+\dfrac{1}{2}x^{\frac{1}{2}}-x^{-\frac{1}{2}}$	M1
	$\mathrm{f}(x)=\dfrac{6x^2}{2}-\dfrac{6}{5}x^{\frac{5}{2}}-\dfrac{1}{3}x^{\frac{3}{2}}+2x^{\frac{1}{2}}(+c)$ or e.g. $\mathrm{f}(x)=\dfrac{6x^2}{2}-\dfrac{3}{5/2}x^{\frac{5}{2}}-\dfrac{1/2}{3/2}x^{\frac{3}{2}}+\dfrac{x^{\frac{1}{2}}}{1/2}(+c)$ or e.g. $\mathrm{f}(x)=\dfrac{6x^2}{2}-\left(\dfrac{3}{5/2}x^{\frac{5}{2}}+\dfrac{1/2}{3/2}x^{\frac{3}{2}}-\dfrac{x^{\frac{1}{2}}}{1/2}\right)(+c)$	M1A1A1
	$12=\dfrac{6(4)^2}{2}-\dfrac{6}{5}(4)^{\frac{5}{2}}-\dfrac{1}{3}(4)^{\frac{3}{2}}+2(4)^{\frac{1}{2}}+c\Rightarrow c=...$	M1
	$(\mathrm{f}(x)=)3x^2-\dfrac{6}{5}x^{\frac{5}{2}}-\dfrac{1}{3}x^{\frac{3}{2}}+2x^{\frac{1}{2}}+\dfrac{16}{15}$	A1
		(6)
		Total 10

11(a)	$x = \frac{5\pi}{2}$ **or** $y = 12$	B1
	$x = \frac{5\pi}{2}$ **and** $y = 12$	B1
		(2)
(b)	$x = \frac{3\pi}{2}$ **or** $y = -21$	B1
	$x = \frac{3\pi}{2}$ **and** $y = -21$	B1
		(2)
(c)(i)	$(A =) -12$	B1
(ii)	$(B =) \frac{5\pi}{4}$	B1
		(2)
		Total 6